

DIGESTION & NUTRITION

Grade Eight Science – Unit 2

MYP GLOBAL CONTEXT & KEY CONCEPTS

Global Context Exploration:

- Scientific and Technical Innovation



Key Concept:

- Relationships

Related Concepts:

- Function
- Consequences

FACTUAL & CONCEPTUAL QUESTIONS

- What molecules do organisms need to function?
- How has scientific understanding been applied to help people get adequate nutrition?



HOW DO OUR DECISIONS AFFECT OUR BODY SYSTEMS?

Respiratory:

- Choosing to smoke or vape
- Pollution
- Choosing to get exercise

Circulatory:

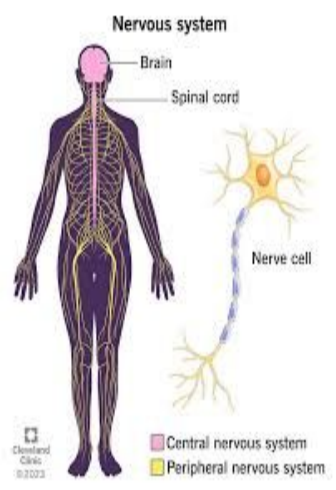
- Choosing to smoke or vape
- The foods we choose to eat
- Choosing to get exercise

Digestive:

- The foods we choose to eat
- Choosing to smoke or vape
- Choosing to get exercise
- Getting enough water
- Choosing to drink alcohol

Nervous:

- What foods we choose to eat
- Protecting ourselves during physical activity
- Choosing to drink alcohol or take drugs



THE NERVOUS SYSTEM

POSITIVE

- **Healthy diet** rich in **B vitamins** and **omega 3** fatty acids (both are important for brain health)
- **Getting enough sleep** – helps to maintain the pathways in the brain so you can learn and create memories

NEGATIVE

- **Alcohol and drugs**– can damage neurons impacting brain chemistry
- **Lack of protection during physical activity** – can risk damaging the spinal cord
- **Poor diet of nutrient poor foods and too much sodium** – can inflame neurons and prevent new neurons from forming. This can impact how the brain works.

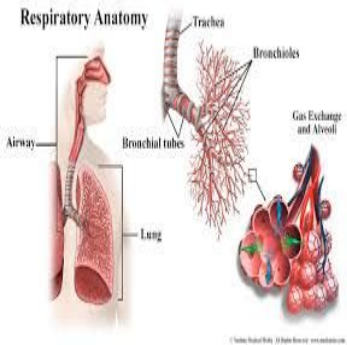
DIGESTIVE SYSTEM CHOICES

POSITIVE

- **Eating a healthy, balanced diet low in saturated fats and sugars** – helps to protect against diseases such as heart disease and diabetes
- **Adequate water intake** – so you don't become dehydrated and constipated

NEGATIVE

- **Eating a poor diet** full of low nutrient foods and a lot of **sodium** – can lead to diseases or other problems such as weight gain and high blood pressure
- **Alcohol** – too much alcohol can damage the liver which is an essential digestive organ to filter toxins out of the body
- **Poor water intake** – can lead to dehydration and constipation



RESPIRATORY SYSTEM CHOICES

POSITIVE

- **Regular exercise** to improve lung capacity and increase stamina
- **Healthy diet**, rich in **Vitamin C** (protects the lungs from infection), and **Vitamin E** (helps maintain normal lung cells and helps to protect lungs from pollution)

NEGATIVE

- **Smoking and vaping** – smoke contains harmful chemicals that can damage lungs and cause cancer. Smoke also causes blood vessels to become more narrow affecting blood pressure
- **Pollution** – can put harmful chemicals into people's lungs

CIRCULATORY SYSTEM CHOICES

POSITIVE

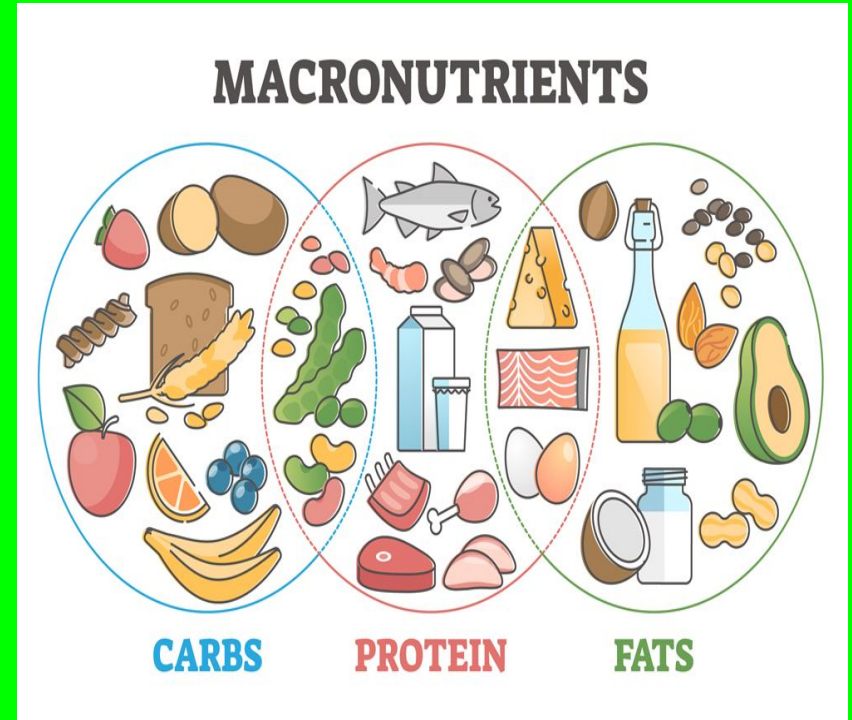
- **Regular exercise** – helps to maintain a healthy weight and build muscle; helps to lower blood pressure, helps to burn off excess fat that can clog arteries
- **Eating a healthy diet** low in **saturated fat** and **rich in healthy fats** such as **omega 3** (helps to reduce blood pressure)
- **Practicing yoga and getting massages** – help to improve blood circulation

NEGATIVE

- **Smoking** – can cause blood vessels to become narrow so blood can't flow as easily
- **Eating an unhealthy diet** high in **saturated fats** and **sodium** – can increase cholesterol and clog arteries; sodium can lead to higher blood pressure by making blood vessels more narrow (harder for the blood to flow through narrow vessels)

FOOD - IT HELPS US LIVE & INTERACT WITH PEOPLE

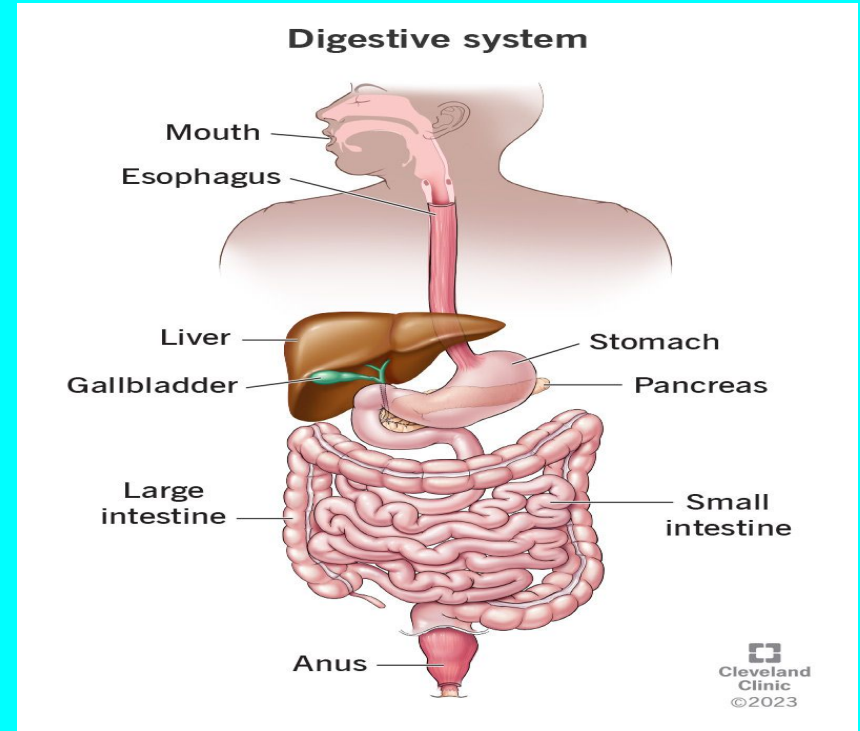
- Food is necessary for life - it is biologically needed for survival
- Food also has important personal, cultural and social connections
- How do we know what foods best fuel our bodies?



<https://www.cedars-sinai.org/blog/what-are-macronutrients.html>

THE DIGESTIVE SYSTEM - A REVIEW FROM BODY SYSTEMS

- The digestive system is the body system that breaks our food into smaller molecules
- It uses physical and chemical changes and turns the food into nutrients that our bodies can use and get rid of waste
- The parts that make up the system are: mouth, pharynx, esophagus, stomach, small and large intestine, rectum and anus.
- The system is helped by the liver, gall bladder, and pancreas



WHAT ARE NUTRIENTS?

- **Nutrients** are the molecules that cells use to perform different functions
- The **nutrient** molecules from our food pass through cell membranes to enter our cells by the process of diffusion
- **RECALL: Diffusion** is the process by which molecules pass from an area of high concentration to an area of low concentration across a semipermeable membrane (RECALL: Nutrients diffuse from the small intestine into the bloodstream)
- Once **nutrient** molecules are inside the cell, cell structures use the molecules to carry out cellular functions and to form different substances to keep the cell alive

THERE ARE TWO TYPES OF NUTRIENTS: MACRO AND MICRO

- **There are two types of nutrients:**

- **Micronutrients** - nutrients needed in relatively small amounts
- **Macronutrients** - nutrients that are needed in large amounts

- **Examples of Micronutrients**

- Iron
- Calcium
- Vitamin A
- Vitamin B
- Vitamin C
- Omega 3
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- **Examples of Macronutrients:**

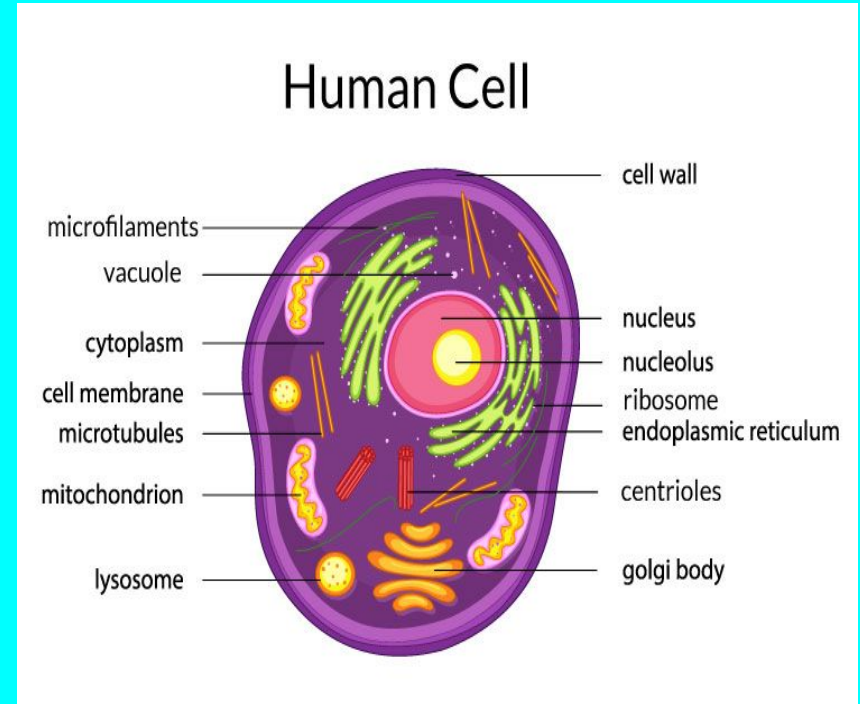
- Fat
- Protein
- Carbohydrates
- Fibre



<https://www.healthcentral.com/nutrition/micros-vs-macros>

OUR CELLS NEED FOOD TO SURVIVE

- **Ribosomes** use substances called amino acids to make proteins
- **Cell membranes** are made of fat, with some protein and carbohydrates
- **DNA, our genetic material inside our cells nucleus**, is made of sugar and protein
- Sugars that have been processed by the cell go into the **mitochondria** and they produce energy
- **Nutrients** from the food we eat pass by **diffusion** into the bloodstream



<https://mytutorsource.com/questions-answers/biology/200/draw-a-diagram-of-a-human-cell-include-the-nucleus-the-golgi-apparatus-lysosomes-ribosomes-and-mitochondria-annotating-your-diagram-with-the-general-functions-of-each-component-and-of-the-cell-membrane/>

WHAT ABOUT FIBRE?

- **Fibre** is a type of **carbohydrate** that our bodies can't fully digest
- **There are two types of fibre** that our bodies can use for different jobs: **soluble fibre** and **insoluble fibre**
- **Soluble fibre** is fibre that our bodies can dissolve
- **Insoluble fibre** is fibre that cannot dissolve in our bodies

The job of soluble fibre: To form a gel like substance to allow the stool to stick together and make it soft and easier to pass

Sources of Soluble Fibre: oats, barley, beans, lentils, fruit, nuts, seeds

The job of insoluble fibre: To add bulk to the stool to make it heavy and easier to pass

Sources of Insoluble Fibre: brown rice, whole wheat flour, quinoa, leafy greens, potatoes (skin on), beans, peas

CALORIES: UNITS OF ENERGY

- The right amount of proteins, fats, and carbohydrates are related to the number of **calories** our bodies need each day
- The number of **calories** needed is different for different people
- **Calories** are units of energy (just like cm is a unit of length and grams is a unit of mass)
- When we talk about **calories** in food, we are referring to the amount of energy the food gives us so we can do the work necessary for survival



(HuLick, 2017)

MACROS: HOW MUCH DO WE NEED & WHY ARE THEY IMPORTANT?

Research tells us that our diets should be made up of about:

- **Carbohydrates** (45% to 65%)
 - **Fats** (20% to 35%)
 - **Proteins** (10% to 35%)
 - **Fibre** (20% to 25%)
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- **Fibre** – helps our stool move; helps us feel more full when eating
 - **Carbohydrates** – break down easily into simple sugars that the cells use to make energy
 - **Fats** – repair and construct cells, make up neurons so signals can be sent quickly around the body, and we need stored fat for long term energy and to keep us warm
 - **Proteins** – growth, production and repair of cells and tissues; special proteins called **enzymes** help with digestion and other body functions

NOT ALL MACRONUTRIENTS ARE CREATED EQUAL

- **Two types of carbohydrates** are called **complex and simple**
- **Complex carbohydrates** are better quality than simple carbohydrates
- It takes longer for the body to break down complex carbohydrates into glucose, which means that the cells produce energy at a slower, more steady pace
- **Simple carbohydrates** break into sugars quickly giving fast bursts of energy but leaving us feeling tired and lacking energy quickly
- **Two types of fat** are called **saturated and unsaturated**
- **Unsaturated fats** are liquid at room temperature. Examples are olive oil, avocado oil, salmon, nuts
- **Saturated fats** are solid at room temperature. Examples are fat from butter, beef, bacon, dark meat of chicken, processed meats like salami and pepperoni

WHAT ABOUT PROTEIN?

- There are different sources of **protein** – **protein from animals** and **protein from plants**
- **Animal Sources:** meat; red and darker meats tend to have more saturated fats than white and lighter colored meats
- Contain all the amino acids we need

- **Plant Sources:** beans, quinoa, whole wheat foods, tofu, soy, nuts and seeds, chickpeas)
- Contain amino acids but don't contain all of amino acids we need



WHAT ABOUT SCIENTIFIC AND TECHNICAL INNOVATION?

What are supplements?

- A substance that contains dietary ingredients
- Examples of dietary supplements are various vitamin and mineral capsules, protein bars, protein powder, and other enhanced foods

Some popular supplements are:

- Probiotics
- Vitamin C
- Omega C
- Vitamin B

Can you think of any other ways that